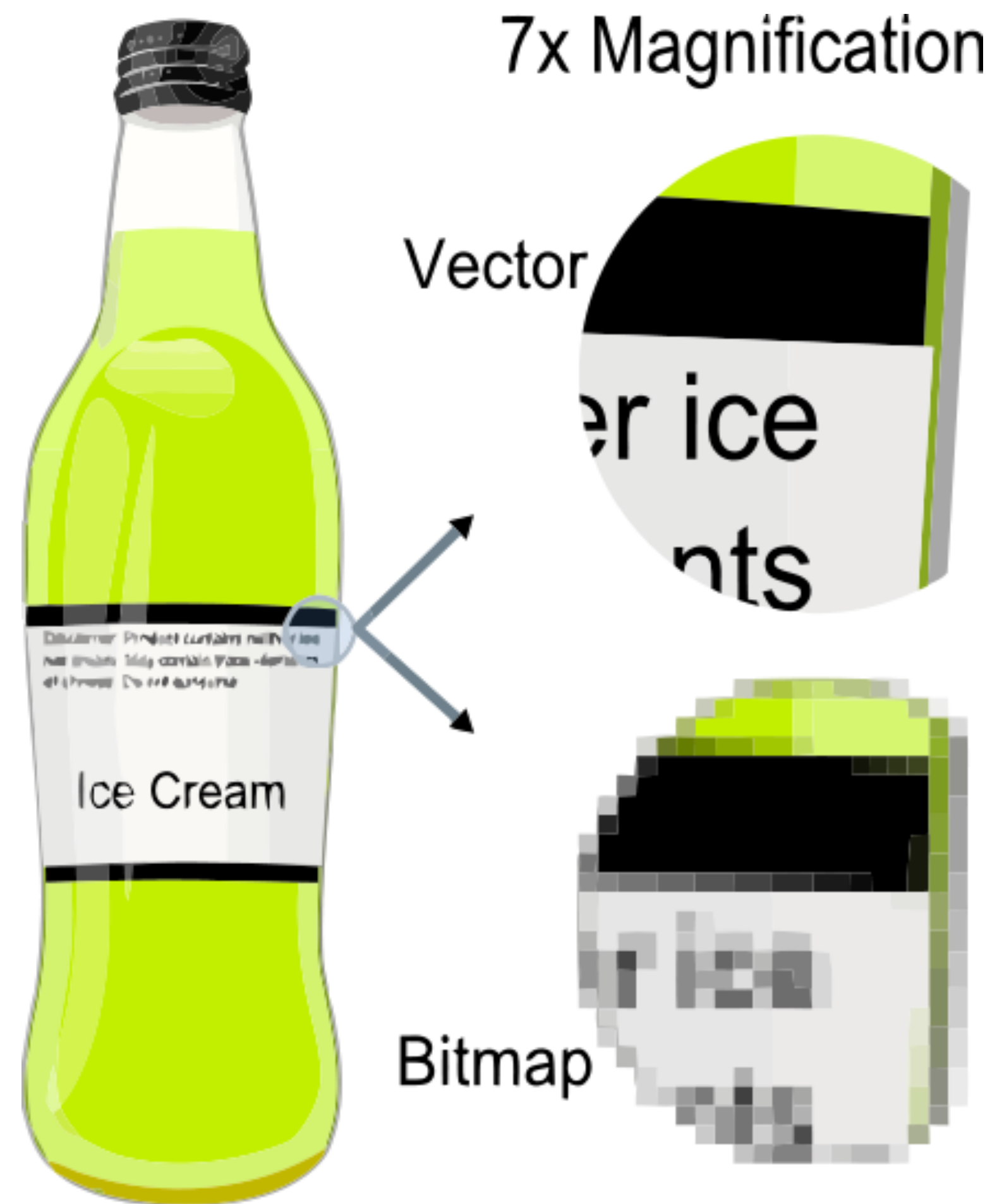


Images

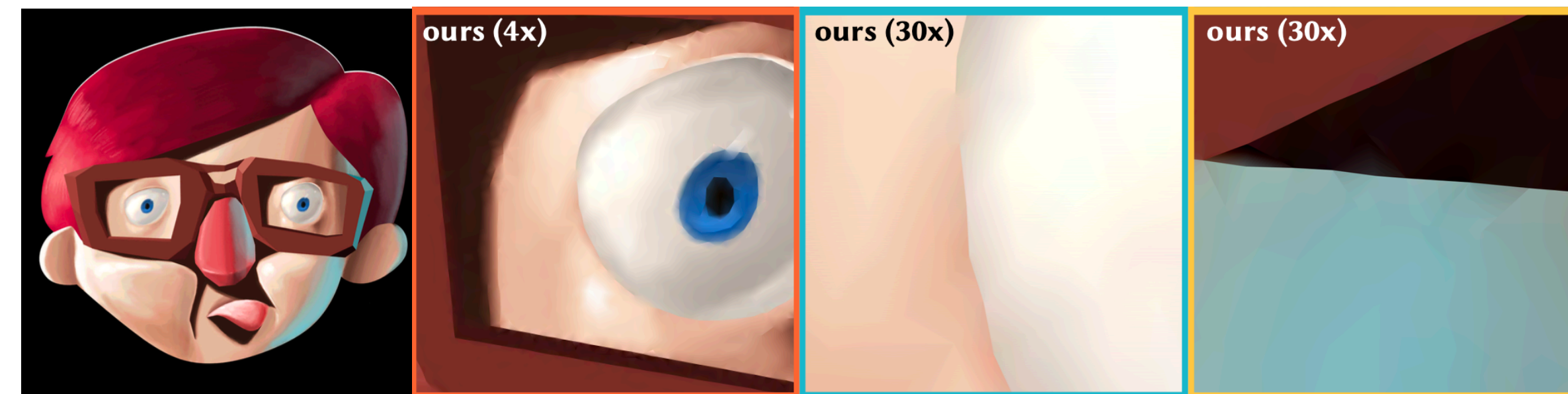
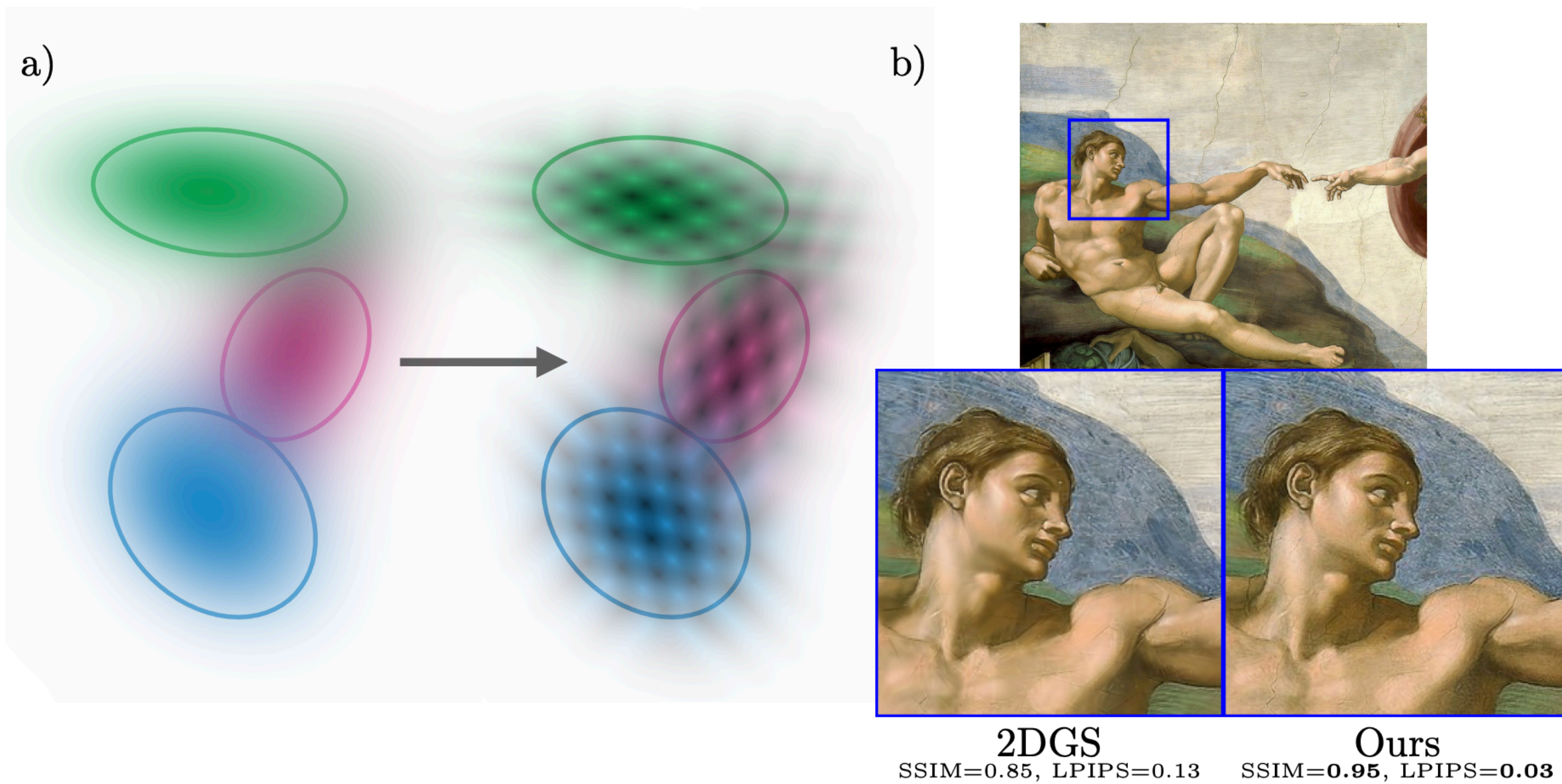


Images



By The original uploader was DARTH Stabro at English Wikipedia - Transferred from en.wikipedia to Commons by Pbroks13 using CommonsHelper., CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=15789788>

Images



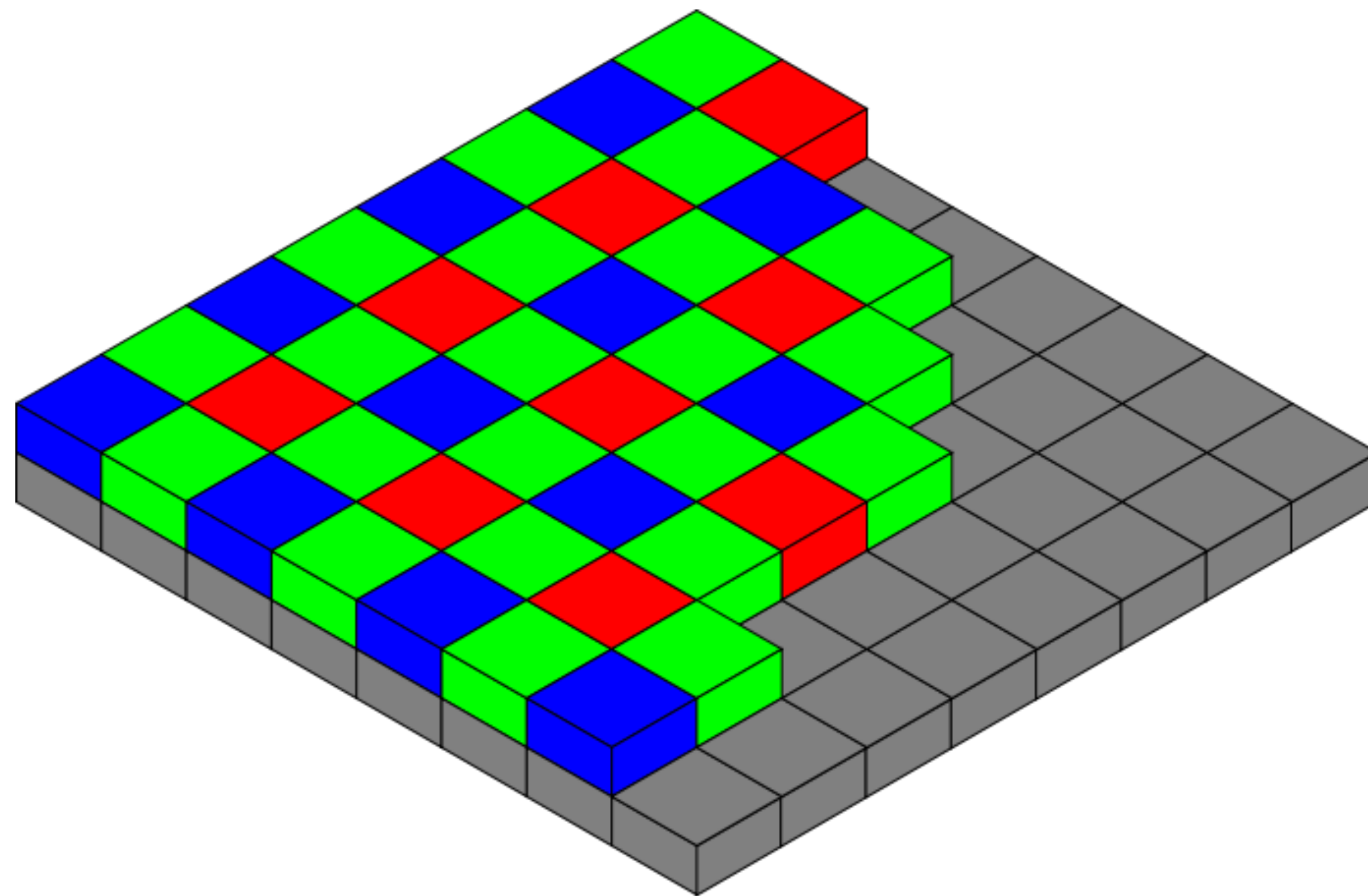
Weiss, Sebastian, and Derek Bradley. "Gaussian billboards: Expressive 2d gaussian splatting with textures." arXiv preprint arXiv:2412.12734 (2024).

Liu, Chenxi, Siqi Wang, Matthew Fisher, Deepali Aneja, and Alec Jacobson. "2D Neural Fields with Learned Discontinuities." In Computer Graphics Forum, vol. 44, no. 2, p. e70023. 2025.

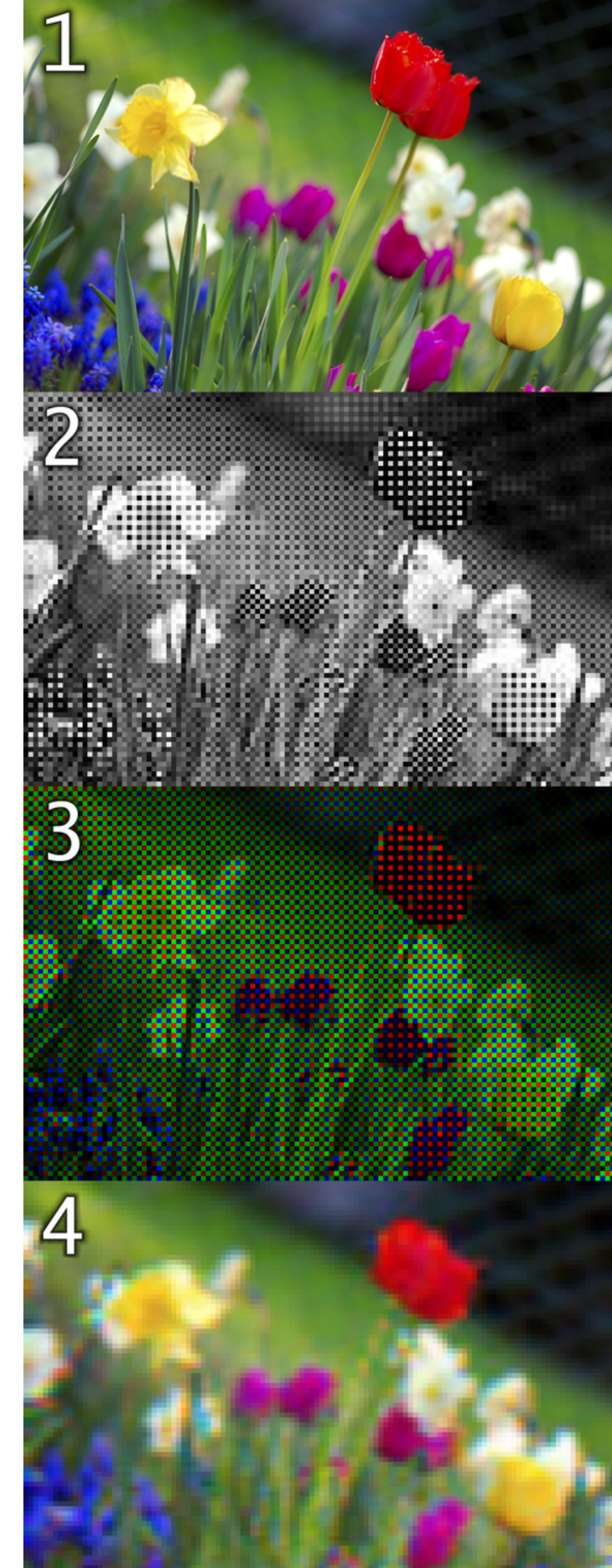
Raster Devices

- Output
 - 2D: Display (LCD,LED)
 - 1D: Hardcopy (ink-jet, dye sublimation)
- Input
 - 2D Array: digital camera
 - 1D Array: scanner

Bayer filter

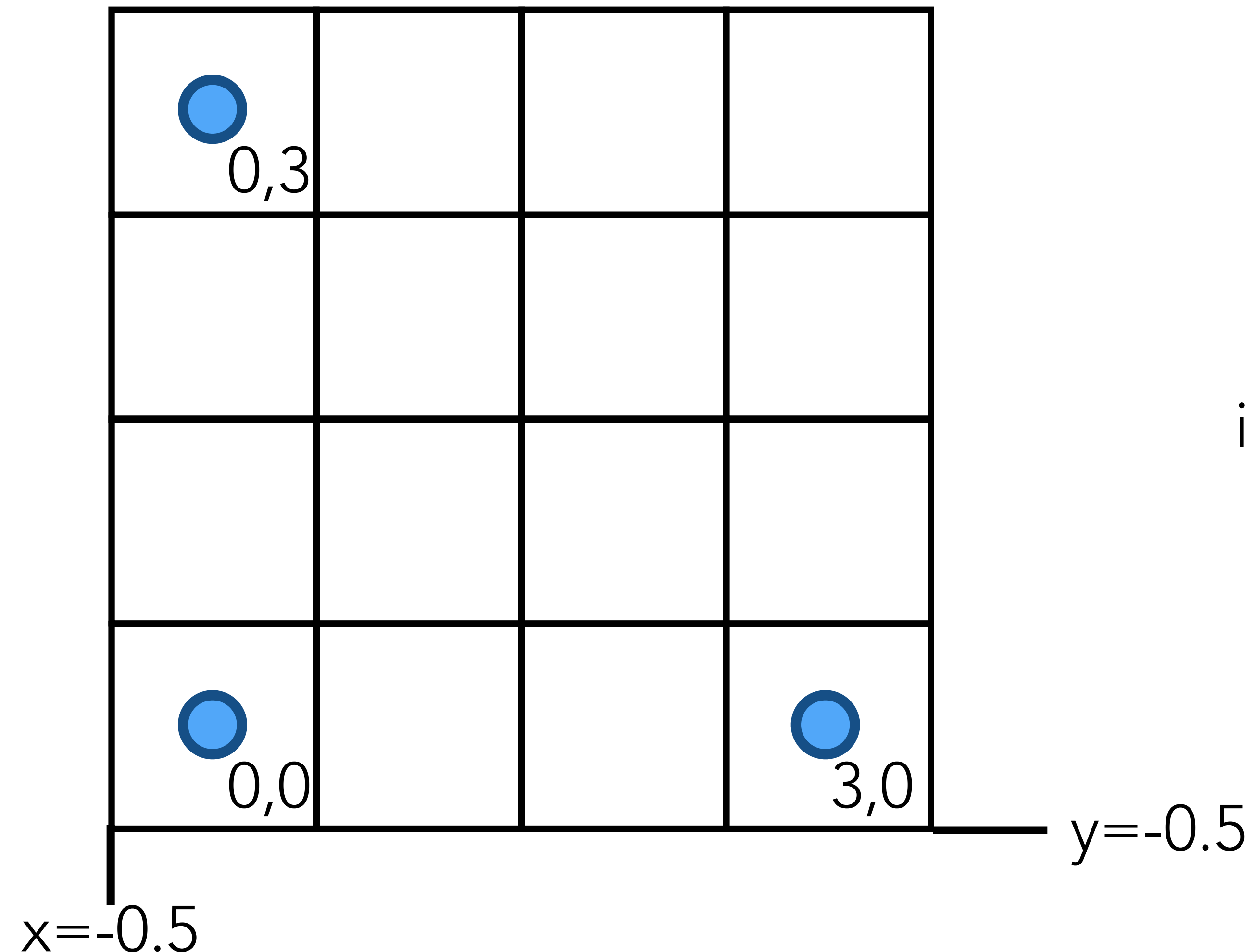


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File:Colorful_spring_garden_Bayer_%2B_RGB.png](https://commons.wikimedia.org/wiki/File:Colorful_spring_garden_Bayer_%2B_RGB.png)

Pixel Coordinates - Raster Image



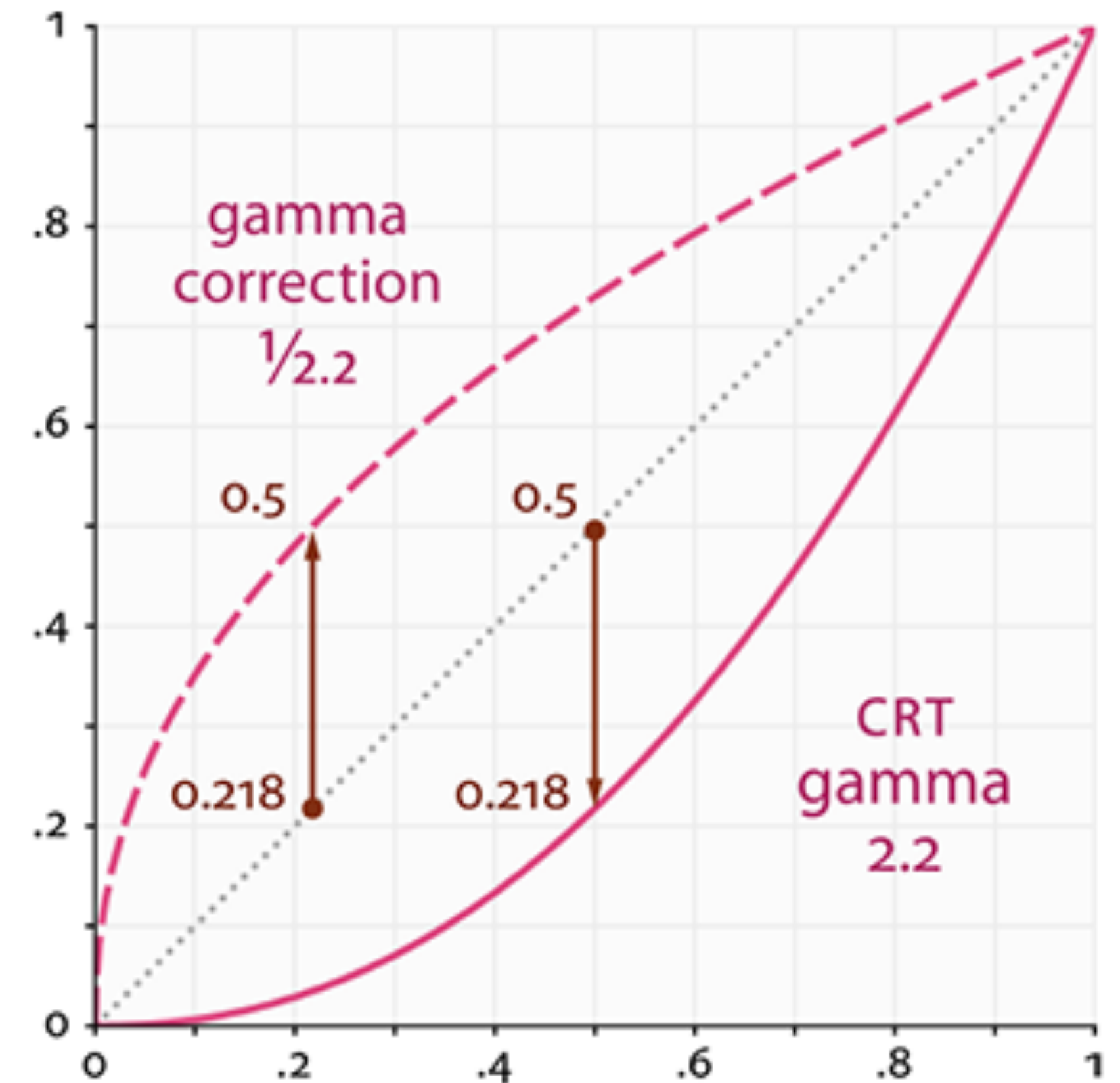
Be Careful:
Y is flipped
in some APIs

Pixel Values (Framebuffer format)

- 1-bit greyscale - text
- 8-bit RGB (24 bits) - web and email
- 8-bit RGBA (32 bits) - alpha channel, see next slide
- 16/24/32bits - high accuracy for photography and HDR

Monitors Intensity, Gamma Correction

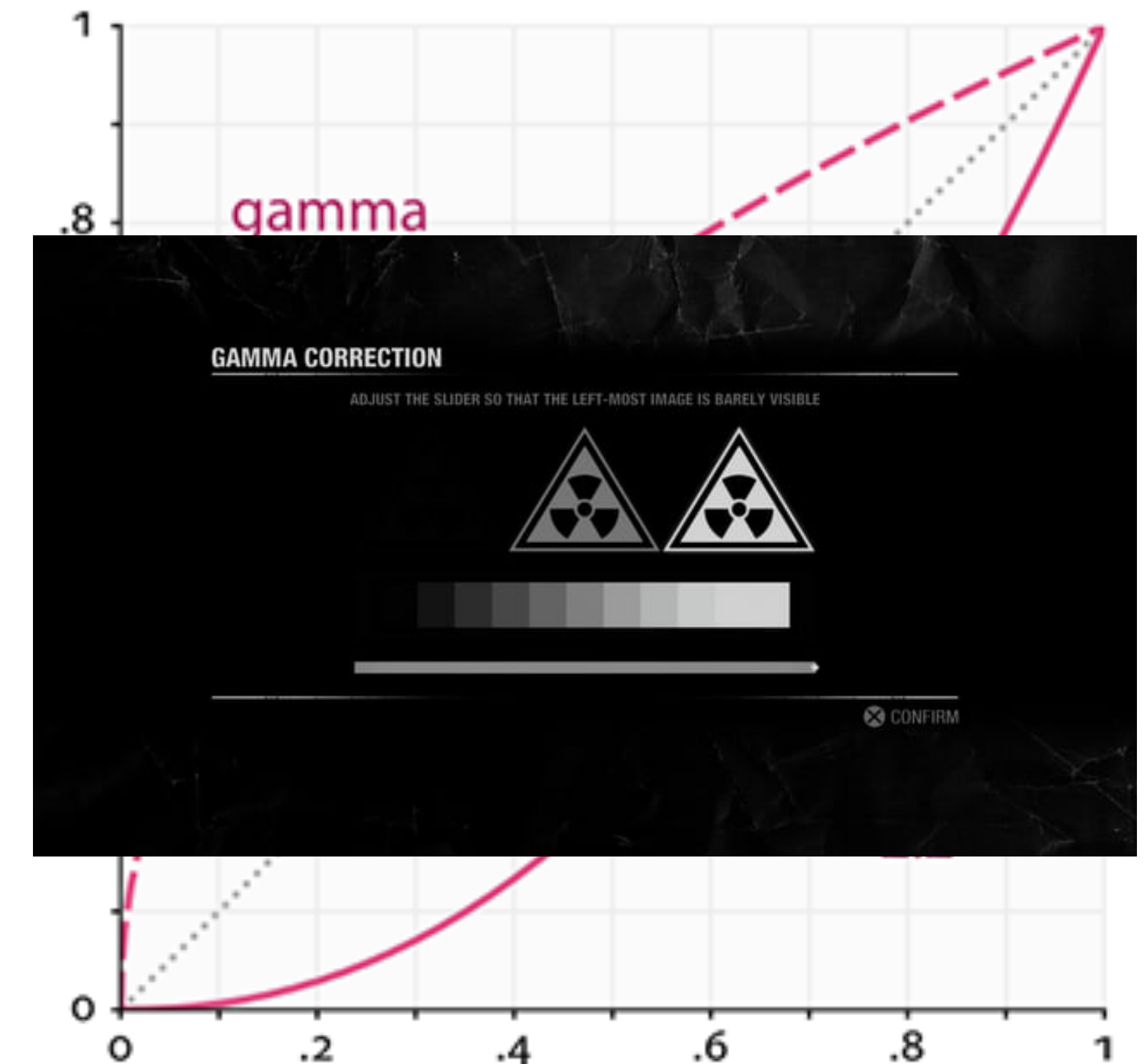
- What is the minimal and maximal light intensity?
- The relationship between pixel value and emitted light is non-linear
- Displays are calibrated for their own response and viewing conditions, by measuring γ



<https://learnopengl.com/Advanced-Lighting/Gamma-Correction>

Monitors Intensity, Gamma Correction

- The display's response is: displayed intensity = $A * v^\gamma$ (A: max intensity, v: pixel value in [0, 1])
- So to display a desired intensity $I = A * a$, we send $v = a^{1/\gamma}$. This is "gamma correction"

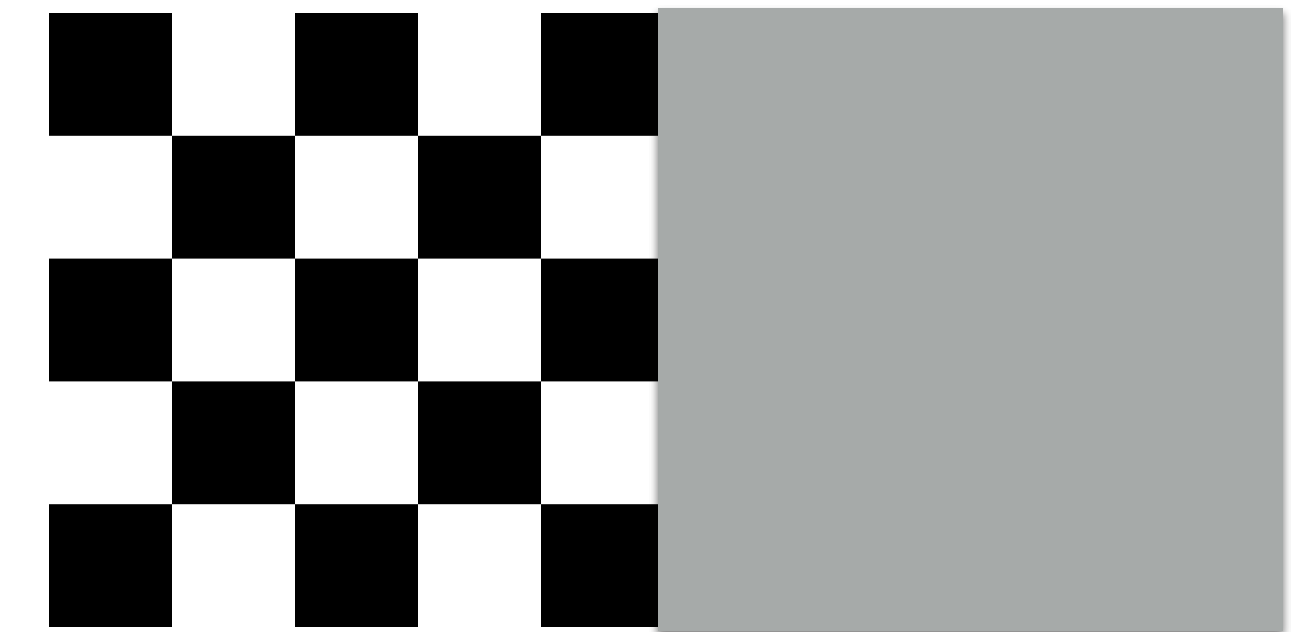


<https://learnopengl.com/Advanced-Lighting/Gamma-Correction>

Gamma Correction



- Find the neutral gray: $0.5 = a^\gamma$
- Compute
$$\gamma = \frac{\ln 0.5}{\ln a}$$



- The colors will not be uniform on normal screens, one of the major factor affecting the cost of screens is their ability to be consistent on all pixels!

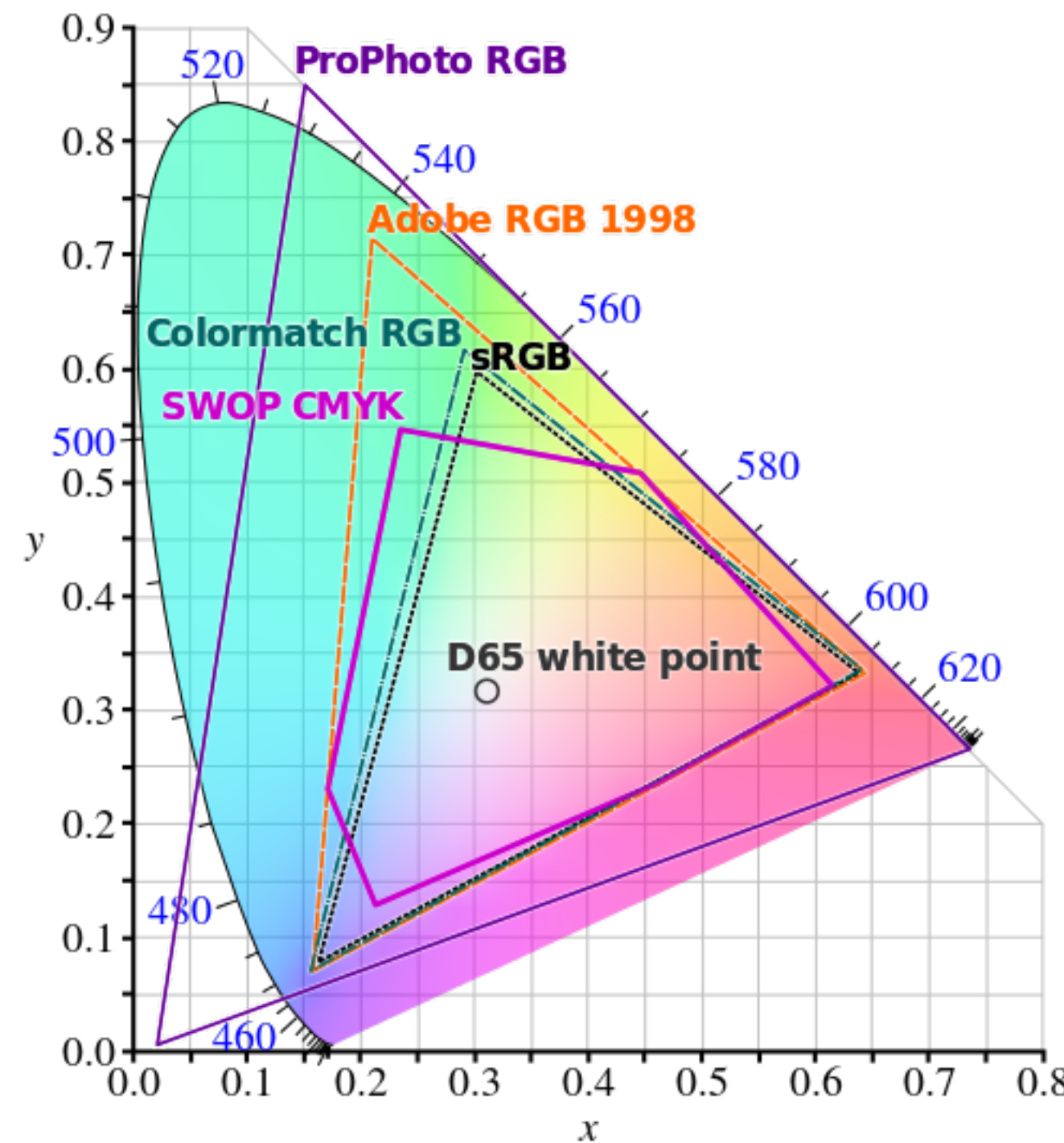
By X-romix 10:00, 7 June 2008 (UTC), Updated by --Rubybrian (talk) 14:25, 14 September 2010 (UTC); Photographer: Toni Frissell - This file was derived from: Weeki Wachee spring 10079u.jpg, GFDL, <https://commons.wikimedia.org/w/index.php?curid=4176109>

RGB vs CMYK colors

- RGB is additive



- CMYK is subtractive

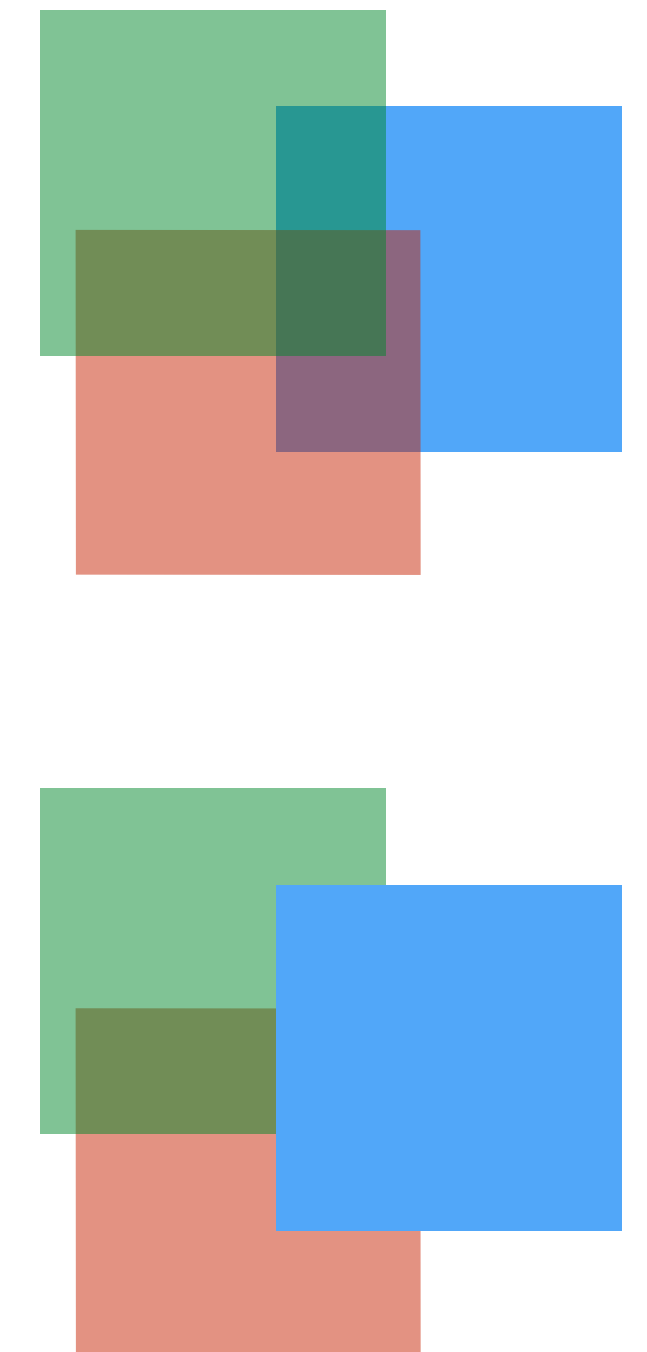
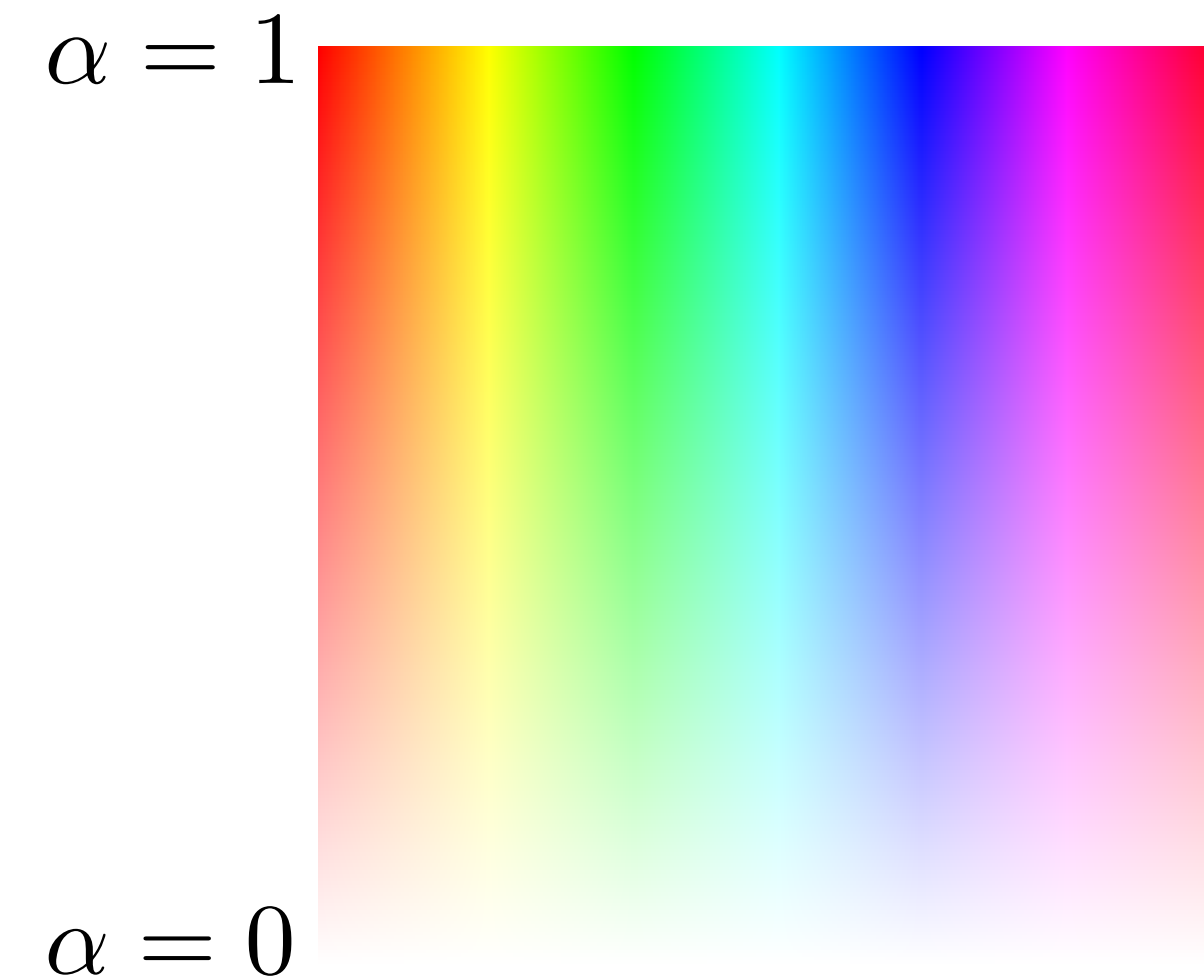


By BenRG and cmglee - http://commons.wikimedia.org/wiki/File:CIE1931xy_blank.svg, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=32158329>

Calibration is very important!

Alpha Compositing

- A way to represent transparency
- The pixels of an image are blended linearly with the image below
- $\mathbf{c} = \alpha \mathbf{c}_{\text{new}} + (1 - \alpha) \mathbf{c}_{\text{old}}$



Ordering matters

RGBA is very common, and you will use it often!

Image Formats

- Lossy:
 - **jpeg** - compact, introduces artifacts
- Lossless:
 - **png** - common for web applications
 - **ppm** - very simple, not compressed
 - **tiff** - mostly scientific use



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[en.wikipedia.org/wiki/
File:Felis_silvestris_silvestris_small_gradual_decrease_of_q](https://en.wikipedia.org/wiki/File:Felis_silvestris_silvestris_small_gradual_decrease_of_quality_-_JPEG_compression.jpg)

[uality_-_JPEG_compression.jpg](https://en.wikipedia.org/wiki/File:Felis_silvestris_silvestris_small_gradual_decrease_of_quality_-_JPEG_compression.jpg)

References

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Chapters 1,2,3